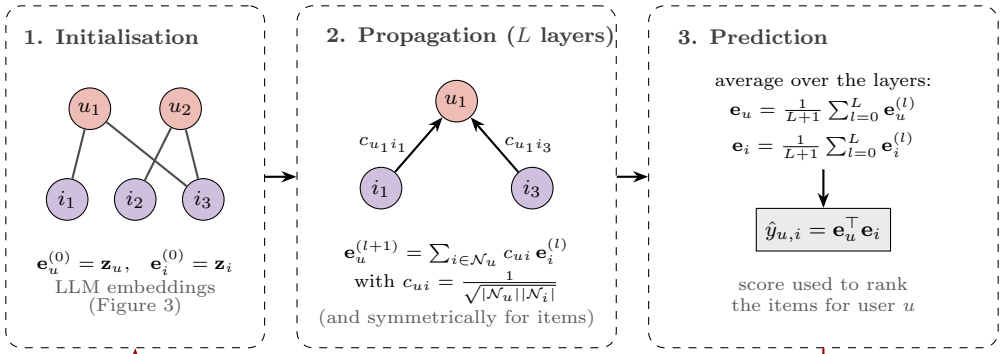

1. Initialisation
u_1
u_2
i_1
i_2
i_3
$\mathbf{e}_u^{(0)} = \mathbf{z}_u, \quad \mathbf{e}_i^{(0)} = \mathbf{z}_i$
LLM embeddings
(Figure 3)
2. Propagation (L layers)
u_1
$c_{u_1 i_1}$
$c_{u_1 i_3}$
i_1
i_3
$\mathbf{e}_u^{(l+1)} = \sum_{i \in \mathcal{N}_u} c_{ui}\, \mathbf{e}_i^{(l)}$
with $c_{ui} = \frac{1}{\sqrt{|\mathcal{N}_u|\,|\mathcal{N}_i|}}$
(and symmetrically for items)
3. Prediction
average over the layers:
$\mathbf{e}_u = \frac{1}{L+1} \sum_{l=0}^{L} \mathbf{e}_u^{(l)}$
$\mathbf{e}_i = \frac{1}{L+1} \sum_{l=0}^{L} \mathbf{e}_i^{(l)}$
$\hat{y}_{u,i} = \mathbf{e}_u^\top \mathbf{e}_i$
score used to rank
the items for user u
training: the BPR loss on triplets (u, i^+, i^-) updates the initial embeddings $\mathbf{E}^{(0)}$